

GIANCARLO POÉMAPE

Machine Learning Engineer | Senior Data Engineer • 12+ Years of Experience
giancarlo.poemape@hotmail.com | [linkedin.com/in/giancarlopoemape](https://www.linkedin.com/in/giancarlopoemape) | Lima, Peru

PROFESSIONAL SUMMARY

Software and Machine Learning Engineer with 12+ years of experience and a Master's degree in Data Science & Artificial Intelligence, specializing in production ML systems, LLM-enabled applications, and large-scale data platforms. Hands-on experience designing and improving ML models, tracking reproducible experiments with MLflow, evaluating RAG and semantic-search systems, and scaling inference for high-volume workloads. Strong background in AWS, Databricks, and Spark, bridging model experimentation with reliable production systems.

TECHNICAL SKILLS

Machine Learning & AI: Model Training & Evaluation, Feature Engineering, Clustering, XGBoost, scikit-learn, PyTorch, TensorFlow, MLflow, Pandas, NumPy

GenAI & LLM Systems: LLMs, RAG, Embeddings, Retrieval Evaluation, LangChain, LangGraph, LangSmith, MCP, Agent-to-Agent (A2A), PGVector, Milvus, OpenSearch, vLLM, ReAct, CoT

Programming & Databases: Python, SQL, PostgreSQL, MySQL, SQL Server, Redshift, Aurora PostgreSQL, JavaScript, TypeScript

Data Engineering: PySpark, Databricks, Delta Lake, Apache Airflow, ETL/ELT, Lakehouse, Data Warehousing, Big Data, JSON/PDF Parsing

Cloud & ML Infrastructure: AWS (Glue, EMR / EMR Serverless, Lambda, Kinesis, SQS, Step Functions, Redshift, Athena, S3, SageMaker, Bedrock, Lake Formation, AppFlow, Cognito, ECS, Fargate), CDK, CloudFormation, Terraform, Docker, Kubernetes, Databricks Asset Bundle

Practices & Tooling: Agile / Scrum, CI/CD, Event-Driven Architecture, Git (GitHub, GitLab, Bitbucket), REST APIs, Metabase

Core Strengths: Troubleshooting, Critical Thinking, Innovation, Service Culture

PROFESSIONAL EXPERIENCE

Senior Data Engineer

Sep 2025 – Present

DolphData - Argentina (Remote)

- Designed and trained a calibrated XGBoost fake-name classifier with MLflow; removed a nearest-neighbor feature responsible for ~99% of lookup latency for a 150M-record daily workload.
- Built a multilingual job-title clustering pipeline using embeddings, evaluating approaches for semantic grouping, taxonomy generation, and production inference.
- Led the migration of data and ML workloads to Databricks (Delta Lake + Spark), improving orchestration, performance, reliability, and cost across production pipelines.

ML Researcher - ProCiencia Project

Nov 2025 – Present (Part-time)

UTEC - Peru

- ML Researcher on DeepIsolation, a state-funded ProCiencia project developing ML/DL pipelines to automate damage classification of seismic isolators using ambient vibration signals.
- Designed and evaluated experimental pipelines combining spectral representations (FFT, CWT, Wavelet Scattering), feature-extraction methods (PCA, 1D Convolutional Autoencoders), and classifiers (SVM, Random Forest, CNN, Vision Transformer), with a strong focus on interpretability.
- Tech stack: MLflow, Python, PyTorch, scikit-learn, Pandas, Matplotlib, Git.

Senior Data Engineer / ML Engineer

Nov 2021 – Jul 2025

Hatchworks - USA (Remote)

- Built LLM and semantic-search solutions using RAG, embeddings, vector databases, and agent architectures; evaluated retrieval quality through precision, recall, and human-in-the-loop review.
- Developed an AI Analyst for PCI DSS compliance using Llama-based models and RAG on AWS SageMaker, within a solution that also incorporated LoRA-adapted Falcon and Llama 2 models.
- Designed and implemented end-to-end data pipelines - lakehouses and warehouses - ensuring data quality, reliability, and scalability, leveraging Amazon EMR / EMR Serverless, Glue, Lambda, Bedrock, AppFlow, Athena, and S3 to streamline data workflows and model deployment.

Data Engineer

Aug 2021 – Jan 2022

Zenta Group - Chile (Remote)

- Built data pipelines with Apache Airflow (Python) to make data available to business areas.
- Developed ETL pipelines and parsers for JSON and PDF files, and delivered on-demand reports in Metabase using Redshift as the database.
- Additional tools: Docker, Kubernetes, AWS.

Cloud Developer & Data Engineer

Feb 2020 – Jul 2021

Taxtech - Peru

- Automated accounting electronic-book processes required by SUNAT using microservices and serverless architecture on AWS (Cognito, ECS, Fargate, S3, CodeCommit, RDS).
- Orchestrated resources with Step Functions and built ETL / Big Data workflows with Glue, Lake Formation, Athena, and PySpark, under the Scrum framework.
- Ingested and transformed hundreds of millions of records using AWK and standard Linux command-line tools, and optimized complex Oracle SQL queries to improve processing performance and resource efficiency.

KEY PROJECTS

Cross-Network Profile Matching System - AWS

- Orchestrated PySpark jobs (EMR / Glue) with Airflow to normalize identities, compute similarity scores, run validations/QA, integrate external APIs, and publish deduplicated audiences into Amazon Redshift for targeted marketing campaigns; close collaboration with business areas was key.

Multi-Agent Medical Prescription System - LLMs

- Designed an LLM-powered multi-agent system to process handwritten medical prescriptions and determine insurance-tier coverage, using A2A architecture with MCP, LangChain/LangGraph, and LangSmith for observability. Built agents for image preprocessing, prescription data extraction, entity matching against a pharmaceutical database, and business-rule enforcement.

PCI DSS Compliance ML Analyst

- Built a PCI DSS compliance analyst using Llama-based models and RAG on AWS; the broader solution also incorporated LoRA-adapted Falcon and Llama 2 models (SageMaker, Glue, S3, OpenAI Embeddings, LangChain, Step Functions, Lambda, Aurora PostgreSQL).

Semantic Search Candidate Recommender

- Developed a semantic search solution to recommend tailored candidates from a role description, using OpenAI Embeddings, PGVector, and OpenSearch as vector databases with PostgreSQL, and Terraform for CI/CD.

CI/CD Monitoring & Alerting Data Pipeline

- Implemented an event-driven monitoring and alerting pipeline on AWS using custom CloudWatch dashboards, CodeCommit, EventBridge, Lambda, CDK, CodeBuild, and CodePipeline.

EDUCATION

Master's Degree, Data Science & Artificial Intelligence

Apr 2024 – Mar 2026

Universidad de Tecnología e Ingeniería (UTEC) - Peru

Bachelor's Degree, Systems Engineering

Mar 2010 – Dec 2014

National University of the Peruvian Amazon (UNAP) - Peru

American English Program

Apr 2019 – 2023

Instituto Cultural Peruano Norteamericano (ICPNA) - Peru

CERTIFICATIONS

- AWS Certified Solutions Architect – Associate - Amazon Web Services (Aug 2022)
- AWS Certified Cloud Practitioner - Amazon Web Services (May 2022)
- Machine Learning Specialization - Coursera & DeepLearning.AI (Jan 2023)
- Unsupervised Learning & Reinforcement Learning - Coursera (Jan 2023)
- TensorFlow Developer Specialization - Coursera & DeepLearning.AI (Dec 2020)
- Big Data Engineer - Big Data Academy, Lima (Mar 2020)